A likelihood ratio test for q-Weibull model selection under interval censoring

Huirong Cao[1], Fuchang Wang[2]
[1] College of Science, Langfang Normal University, Langfang, China
[2] Emergency Management University, Beijing, China

Abstract

The q-Weibull distribution extends the Weibull by introducing a shape deformation parameter q that controls tail heaviness ($q > 1$) or bounded support ($q < 1$). Under interval censoring, estimating q is challenging because the tail information is partially observed. We propose a likelihood ratio test for comparing q-Weibull against Weibull, and show via simulation that the test has near-nominal size and reasonable power. We apply the test to a large dental emergence study (n=4386) and find significant evidence against the Weibull in favor of the q-Weibull with $q > 1$, consistent with unobserved heterogeneity.

1 Introduction

The Weibull distribution is the standard parametric model in survival analysis and reliability engineering. Its survival function is $S(t) = \exp(-\lambda t^{\beta})$. However, the Weibull has a fixed exponential tail and unbounded support. Real lifetime data often exhibit either heavier tails (due to unobserved heterogeneity) or bounded maximum lifetimes.

The q-Weibull distribution (**?**) introduces a deformation parameter q:

$$S_q(t) = \left[1 + (q-1)\lambda t^{\beta}\right]^{-1/(q-1)}, \quad q \neq 1, \tag{1}$$

which reduces to the Weibull as $q \to 1$. When $q > 1$, the tail decays as a power law (heavier than Weibull); when $q < 1$, the distribution has a finite upper endpoint $T_{\max} = [1/((1-q)\lambda)]^{1/\beta}$.

Interval-censored data arise when the exact event time is unknown, but it is known to lie in an interval $(L_i, R_i]$. This is common in periodic follow-up studies. Under interval censoring, the tail of the distribution is less precisely identified, making the estimation of q particularly difficult.

We propose a likelihood ratio test for $H_0 : q = 1$ (Weibull) vs $H_1 : q \neq 1$ (q-Weibull), under interval censoring. We evaluate the test via simulation and apply it to a large dental dataset.

2 Methods

2.1 Model

For interval-censored observations $(L_i, R_i]$, the likelihood contribution is $S(L_i) - S(R_i)$. For right-censored observations at L_i, the contribution is $S(L_i)$.

We fit both the Weibull ($q = 1$) and q-Weibull (free q) by maximum likelihood. The likelihood ratio statistic is:

$$\Lambda = 2\left[\ell(\hat{q}, \hat{\lambda}, \hat{\beta}) - \ell(q = 1, \tilde{\lambda}, \tilde{\beta})\right]. \tag{2}$$

Under H_0, Λ is approximately χ^2_1. However, because interval censoring reduces tail information, the asymptotic approximation may be poor in small samples. We also provide a parametric bootstrap alternative.

2.2 Interpretation

- If $q < 1$ is supported: the data suggest bounded support. - If $q > 1$ is supported: the data suggest a heavier-than-Weibull tail. Note that this can arise from either genuine tail deformation or unobserved heterogeneity (frailty), which produces the same survival function form (**?**).

3 Simulation study

We generate interval-censored data mimicking annual examinations: $L = \lfloor t \rfloor$, $R = L + 1$, with 37% right censoring. We set $\lambda = 1/16$, $\beta = 2$, and vary q. We run 200 replications per scenario and use the asymptotic χ^2_1 reference distribution.

Table 1: Simulation results (200 reps). Size under Weibull null.

Scenario	$\hat{q}$ (mean±sd)	Λ (mean)	Rejection rate ($\alpha = 0.05$)
Weibull, n=200	0.93 ± 0.28	1.13	0.070
q=0.8, n=200	0.56 ± 0.29	2.59	0.255
q=0.7, n=200	0.46 ± 0.25	3.57	0.365
q=0.6, n=200	0.34 ± 0.21	4.84	0.580
Weibull, n=500	0.94 ± 0.16	0.99	0.030
q=0.7, n=500	0.41 ± 0.19	8.70	0.840

The size is close to the nominal 5% (7.0% at n=200, 3.0% at n=500). Power increases with sample size and with deviation from $q = 1$. At n=500 and q=0.7, power reaches 84%.

4 Real data application

We apply the test to the Signal Tandmobiel dental emergence study (n=4386), where children were examined annually and tooth emergence times are interval-censored.

Table 2: Real data results.

Model	nll	Parameters
Weibull	5597.0	λ, β
q-Weibull	5576.7	q, λ, β
LR statistic	$\Lambda = 40.6$, $p < 0.0001$	
Estimated q	$\hat{q} = 1.35$	

The test strongly rejects the Weibull in favor of the q-Weibull. The estimated $q = 1.35$ indicates a heavier-than-Weibull tail. As noted above, this is consistent with either genuine tail deformation or unobserved heterogeneity (frailty). The q-Weibull survival function with $q > 1$ is mathematically equivalent to a gamma frailty model with $\theta = 1/(q-1) \approx 2.86$.

5 Discussion

We have proposed a likelihood ratio test for q-Weibull model selection under interval censoring. The test has near-nominal size and reasonable power in moderate samples. The real data application demonstrates that the test can detect deviations from the Weibull assumption.

However, two caveats are important. First, a significant test in the direction $q > 1$ does not distinguish between tail deformation and frailty. Second, when $q < 1$, the test detects bounded support, but estimating the exact upper endpoint remains difficult under interval censoring.

Future work will develop a formal test for distinguishing $q > 1$ tail deformation from frailty, perhaps using additional model features such as the hazard function shape.

References

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